

15th annual International Conference on Education and New Learning Technologies



Analysing the effectiveness and outcomes of
blended and self-directed *e-learning* approach
through the MOODLE platform for statistical training
in Social Sciences degrees

M. Pérez Priego M^a B. García-Moreno G. Gómez-Casero
J. R. Caro Barrera

Palma de Mallorca, 3rd – 5th of July, 2023

Índice

- 1 Introduction
 - The University of the 21st century
 - The new teaching methodologies
 - Our research
- 2 Design and Methodology
 - Developed Activities
 - Data Collection and Processing
 - Quantitative results
- 3 Analysis and Discussion
 - E-learning activities
 - Tests and final exam
- 4 Conclusions
- 5 References

The 'new' European university framework

Standardisation of the European Higher Education Area (EHEA)

- Improvement of training and educational processes that entails a homogenisation of curricula and training itineraries at the European level.
- Adaptation of the educational offer to the demands of an increasingly demanding labour market.
- Change in the approach to teaching $\Rightarrow$ use of technology and communication:
 - *redesigning the way of learning and teaching.*

$\Rightarrow$ Constant change in the university teacher and student profile.

$\Rightarrow$ **Bologna motto:** “*learning to learn*”. It must be applicable to both teachers and students.

The “*evolution*” of the continuous assessment

The daily monitoring of the student through tutorials and the use of all the possibilities offered by ICTs have evolved the concept of **continuous assessment**.

The university is no longer a strictly face-to-face system, but the new university teaching and learning methodologies are developed in one of the following ways:

Learning modes

- (a) **Pure face-to-face learning systems:** less and less.
- (b) **Pure *online* teaching systems:** *e-learning*.
- (c) **Mixed systems:** a mixture of the two previous ones and with multiple sub-modalities: *blended learning*
 - Virtual lectures.
 - Face-to-face tutorials.

Teaching in a pandemic context

• **ACADEMIC YEAR 2019/2020**

⇒ Official statement of the pandemic (WHO: 12 March 2020).

⇒ Adoption of 100% online teaching, for confinement reasons at the University of Córdoba:

- Move from face-to-face to online teaching quickly (in days/hours) with...
- ...little room for adopting an efficient, flexible and effective methodology, which meant a...
- ...disruption for the teachers: speed in adapting the methodology to the new situation, and
- for students, due to the change in the new way of learning and assessment.

Teaching in a pandemic context

• **ACADEMIC YEAR: 2020/2021**

Summer with mobility restrictions. Spikes in cases after the summer period. Government guidelines were still vague. Beginning of the academic year on a blended learning basis (50% classroom capacity, some students in class, others at home, etc...).

Our analysis:

- Follow-up 20/21 academic year: set in the context of continuous assessment and blended learning in Business Administration, Tourism and Labour Relations/Human Resources.
- Historic low pass rate in Statistics subjects in social sciences degrees.
- Academic year 19/20: increase to levels in line with the training received in the baccalaureate.

1. Developed Activities

The proposed analysis consisted of the following activities:

1. Tests

- 10 questionnaires in the *Moodle*® platform.
- Questions on topics in the specified subject area.
- Correction and assessment on *Moodle*® platform.
- **Additional criteria:** possibility to see the time spent on the questionnaire:
 - Solved in *less than 3 minutes*: NEGATIVE TRIAL.
 - Solved in *between 3 and 10 minutes*: NEUTRAL TRIAL.
 - Solved in *more than 10 minutes*: POSITIVE TRIAL.

2. Lessons

- Complementary tool to the questionnaires.
- Gradual difficulty of the questions: if a mistake was made, the system proposed another question of similar difficulty.
- If a correct answer was given, the system proposed a question with a higher degree of difficulty or from the next part.
- If three questions in a row were not answered correctly, the lesson ended and the system displayed the level acquired and the parts on which to focus the study.

3. Partial tests

- 3 partial tests, one for each block of the subject.
- Did not eliminate subject matter for the final exam
- Did not directly influence the final grade of the subject.

2. Data Collection and Processing

Data Collection from the following sources:

- Subject grades.
- Grades of the mid-term tests.
- Grades of the tests.
- Information on student performance via *Moodle*® platform.

The data were exported to a table reflecting:

- 1) Subject grades.
- 2) Grades of the mid-term tests, as well as an average of them.
- 3) Grades of the midterm tests, as well as an average of them.
- 4) Number of questionnaires solved by each student, detailing the number of positive, neutral and negative trials.

3. Quantitative results

Target:

To detect if there is a correlation between the grades of the proposed questionnaires, or the partial controls, with the final grade of the subject.

Sample:

Total sample of 150 students, after eliminating students who did not take the final exam, even though they had completed the previous activities.

• 3.1. Relationship between tests and final grade

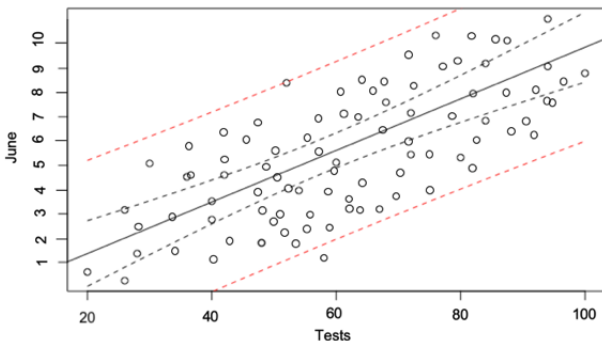


Figure 1. Relationship between test scores and final grades.

$R^2 = 0.8373$: direct and strong relationship between the test results and the final grade, i.e. the higher the score on the questionnaires, the higher the student's grade on the final exam.

• 3.2. *Relationship between the number of tests completed and the final grade:*

Logical reasoning: assume that the more participation in the proposed activities, the higher the grade obtained in the final exam.

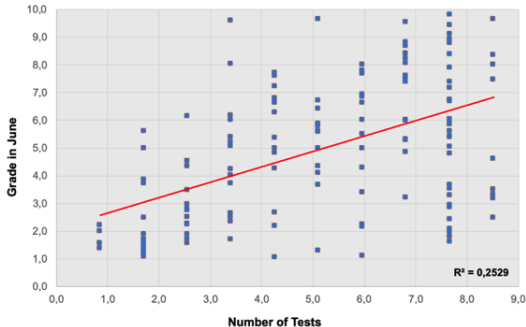


Figure 2. *Relationship between the number of questionnaires and the final grades.*

However $\Rightarrow R^2 = 0.2529$: Weak relationship between the variables.

• 3.3. *Relationship between controls and final grade:*

Study of the correlation between the partial tests and the final grade:

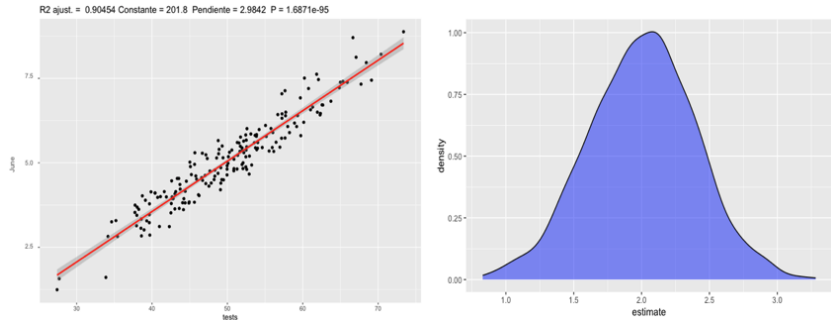


Figure 3. *Correlation between controls and final mark.*

$\Rightarrow R^2 = 0.9045$: Strong relationship between the variable partial controls and the final mark obtained. A high average grade in the tests tends to imply a high mark on the final exam.

Actividades e-learning

Deductions

- **Null correlation in students with ‘good’ attempts:** the sample of students with good attempts is significantly small compared to the total.
- **Too many students with too short attempts:** high reduction of the sample of students with good attempts.
Suspicious that students have taken the questionnaires in groups, spending the right amount of time on one questionnaire and only a few minutes on the others once they have had the questions and answers.
- **Not being able to control the authorship of the questionnaires and lessons,** means that the mark obtained cannot be considered an important part of the final mark.

- **The proposed *e-learning* activities** stimulate and promote competences such as leadership and group work.
- The fact of completing the questionnaires shows the **degree of interest shown by the students in passing the course**, which is reflected in the final marks. This conclusion is reinforced by the weak relationship between the number of questionnaires completed and the final grade.

Tests and final exam

Deductions

- Logical to think that the student who successfully passes the mid-term tests throughout the course is better prepared to face the final exam. Even if they do not pass them, they serve as training for the student and as a monitoring tool for the teacher, indicating the state of the class and its evolution throughout the course.
- As the relationship between the variables is moderately strong, there is a high correlation between the two grades, suggesting the possibility of eventually replacing the final exam with the mid-term tests.

Conclusions and working lines

The proposed method of continuous semi-face-to-face assessment is satisfactory and improves on the method used before the pandemic.

A balanced combination of *e-learning* activities and partial face-to-face assessments reflects the achievement of important competencies, as well as the passing of the statistics subject in the degrees where the initiative has been carried out.

In the current academic year, we are continuing to test and extend the proposed methodology, within the Moodle platform within the *Moodle*® platform, we are using the SCORM module, designing a package that includes web pages, graphics and *Flash*® presentations so that they can be used in any web browser.

We are innovating with the incorporation of new variants with the aim of extending the methodology to other degrees in other branches in order to continue improving the level of the students and, as a consequence, increase the success rate in the subject of statistics.

References

- Aranda, C., Medina Moreno, J. (2012). Influencia de las actividades e-learning de MOODLE en la formación en matemáticas de los alumnos de Grado. En Santamaría Lancho, M. y Sánchez-Elvira Paniagua, A. (Coord.) *I Jornadas Internacionales de Innovación Docente Universitaria en Entornos de Aprendizaje Enriquecidos*. Libro de Actas, UNED, Madrid, Septiembre 2012.
- Molina, I. (2012). Los nuevos retos del profesor universitario. Suplemento ABC Empresas. págs. 34–35.
- Bajo Sanjuán, A. (2010). El potencial de las nuevas tecnologías para el mundo académico. En Bajo Sanjuán, A. y Vilagra García, N. *La ética empresarial y la responsabilidad social en el nuevo contexto digital*. Cátedra Javier Benjumea (Focus-Abengoa), Memoria académica del Curso 2009-2010, Universidad Pontificia de Comillas, Madrid, pág. 169.
- Universidad de Córdoba (2011). Facultad de Ciencias Económicas. Doc. verifica Grado de Administración y Dirección de Empresas (Aneca). Available in: <http://www.uco.es/grados/grado-en-administracion-y-direccion-de-empresas>.
- Universidad de Córdoba (2011). Facultad de Ciencias del Trabajo. Doc. verifica Grado de Turismo. (Aneca). Available in: <http://www.uco.es/docencia/grados/grado-en-turismo>.
- Universidad de Córdoba (2015). Facultad de Ciencias del Trabajo. Doble Grado de Relaciones Laborales y Recursos Humanos. Available in: <http://www.uco.es/docencia/grados/grado-en-relaciones-laborales-y-recursos-humanos>.

Thanks for your attention.